**Airline Travel Insurance Testing**

**Application Under Test: BlazeDemo (demo environment used to simulate airline booking flow such as AirAsia)**

**Objective**

To verify that the travel insurance offering is correctly integrated into the flight booking flow and customers can successfully purchase, modify and receive travel insurance coverage when booking airline tickets.

**1. Manual Test Scenarios / Test Cases**

**Module A: Insurance Display**

**TC001 - Verify Travel Insurance Option Display**

**Precondition:** User searches for a flight.

**Steps:**

1. Select departure and arrival destination.
2. Select travel date.
3. Click Search Flight.
4. Proceed to booking page.

**Expected Result:**

- Travel insurance option is displayed.
- Insurance details are visible.
- Insurance premium amount is shown.

**TC002 - Verify Insurance Terms & Conditions**

**Steps:**

1. Click Insurance Terms & Conditions link.

**Expected Result:**

- Terms and conditions page opens successfully.
- Content is readable.
- No broken links.

**Module B: Insurance Selection**

**TC003 - Purchase Booking With Insurance**

**Steps:**

1. Select travel insurance.
2. Continue booking process.
3. Complete payment.

**Expected Result:**

- Insurance charge is added to total fare.
- Booking succeeds.
- Insurance is included in booking summary.

**TC004 - Purchase Booking Without Insurance**

**Steps:**

1. Do not select insurance.
2. Complete booking.

**Expected Result:**

- Booking succeeds.
- No insurance charges added.

**TC005 - Toggle Insurance Selection**

**Steps:**

1. Select insurance.
2. Verify price update.
3. Unselect insurance.

**Expected Result:**

- Insurance charge removed immediately.
- Total amount recalculated correctly.

**Module C: Price Validation**

**TC006 - Verify Insurance Premium Calculation**

**Steps:**

1. Select flight.
2. Select insurance.

**Expected Result:**

- Insurance premium matches configured value.
- Total amount = Flight Fare + Insurance Premium.

**TC007 - Verify Multiple Passenger Insurance Pricing**

**Steps:**

1. Add multiple passengers.
2. Select insurance.

**Expected Result:**

- Insurance premium calculated correctly for all passengers.

**Module D: Payment Integration**

**TC008 - Successful Payment With Insurance**

**Steps:**

1. Select insurance.
2. Make successful payment.

**Expected Result:**

- Booking confirmed.
- Insurance policy generated.

**TC009 - Failed Payment With Insurance**

**Steps:**

1. Select insurance.
2. Simulate payment failure.

**Expected Result:**

- Booking not confirmed.
- Insurance policy not generated.

**Module E: Booking Confirmation**

**TC010 - Verify Booking Confirmation Page**

**Expected Result:**

- Insurance details displayed.
- Policy reference number displayed.

**TC011 - Verify Confirmation Email**

**Expected Result:**

- Email received.
- Insurance information included.
- Correct passenger details shown.

**Module F: Negative Testing**

**TC012 - Session Timeout During Booking**

**Expected Result:**

- Appropriate error message displayed.
- No incomplete insurance purchase created.

**TC013 - Browser Refresh During Insurance Selection**

**Expected Result:**

- Insurance selection retained correctly.
- No duplicate insurance charge.

**TC014 - Invalid Passenger Information**

**Expected Result:**

- Validation message displayed.
- User cannot proceed.

**Module G: Cross Browser Testing**

**TC015 - Verify Insurance Purchase Across Browsers**

Browsers:

- Chrome
- Edge
- Firefox
- Safari

**Expected Result:**

- Insurance flow works consistently.

## Scenario Chosen for Automation

### Automated Scenario

The selected automation scenario is an End-to-End Flight Booking Flow with Travel Insurance validation.

Since real airline portals have restricted access, dynamic content and payment security controls, a demo application was used to demonstrate the automation approach.

### Test Application Used

BlazeDemo was used as the test application: https://blazedemo.com

### Automated Flow

1. Launch application
2. Select departure and destination cities
3. Search available flights
4. Select a flight
5. Enter passenger details
6. Proceed to purchase
7. Verify booking confirmation message

### Mapping to Real Business Scenario

This flow simulates a real airline booking system where:

- Flight selection = Airline ticket selection
- Passenger details = Customer booking information
- Purchase flow = Payment integration
- Confirmation page = Booking + Insurance confirmation

### Why This Scenario Was Automated

This scenario was chosen because it:

- Represents a critical end to end business flow
- Covers multiple UI components and user interactions
- Suitable for regression testing
- Demonstrates framework design and automation capability
- Can be reused for future test expansions

## 3. Automation Framework Recommendation

### Framework

Katalon Studio (Built on Selenium WebDriver)

Why Katalon Studio

Katalon Studio was selected because it provides a fast and maintainable automation solution while leveraging Selenium WebDriver capabilities.

Advantages:

- Easy test case development and maintenance.
- Supports object repository management.
- Built-in reporting and screenshots.
- Supports data-driven testing.
- Supports Test Suites and Test Suite Collections.
- Integrates with GitHub and CI/CD pipelines.
- Suitable for both web and mobile automation testing.

Automation Proof of Concept

A proof-of-concept automation script was developed using Katalon Studio and published to a public GitHub repository.

The automated scenario covers:

- Flight search
- Flight selection
- Passenger information entry
- Booking submission
- Booking confirmation validation

**Scope of Automation**

- UI flow automation only
- Excludes payment gateway actual processing
- Uses mock/demo data

**Automation Tool Stack**

- Katalon Studio
- Selenium WebDriver
- Groovy scripting
- TestNG style execution (Katalon engine)